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[Submitted on 28 Feb 2021]

MirrorME: Implementation of an IoT based Smart Mirror through Facial Recognition and Personalized Information Recommendation Algorithm

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[Ayesha Siddika Mukta](#), [Uzzal Kumar Acharjee](#)

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We are living in the era of the fourth industrial revolution, which also treated as 4IR or Industry 4.0. Generally, 4IR considered as the mixture of robotics, artificial intelligence (AI), quantum computing, the Internet of Things (IoT) and other frontier technologies. It is obvious that nowadays a plethora of smart devices is providing services to make the daily life of human easier. However, in the morning most people around the globe use a traditional mirror while preparing themselves for daily task. The aim is to build a low-cost intelligent mirror system that can display a variety of details based on user recommendations. Therefore, in this article, Internet of Things (IoT) and AI-based smart mirror is introduced that will support the users to receive the necessary daily update of weather information, date, time, calendar, to-do list, updated news headlines, traffic updates, COVID-19 cases status and so on. Moreover, a face detection method also implemented with the smart mirror to construct the architecture more secure. Our proposed MirrorME application provides a success rate of nearly 87% in interacting with the features of face recognition and voice input. The mirror is capable of delivering multimedia facilities while maintaining high levels of security within the device.

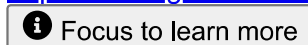
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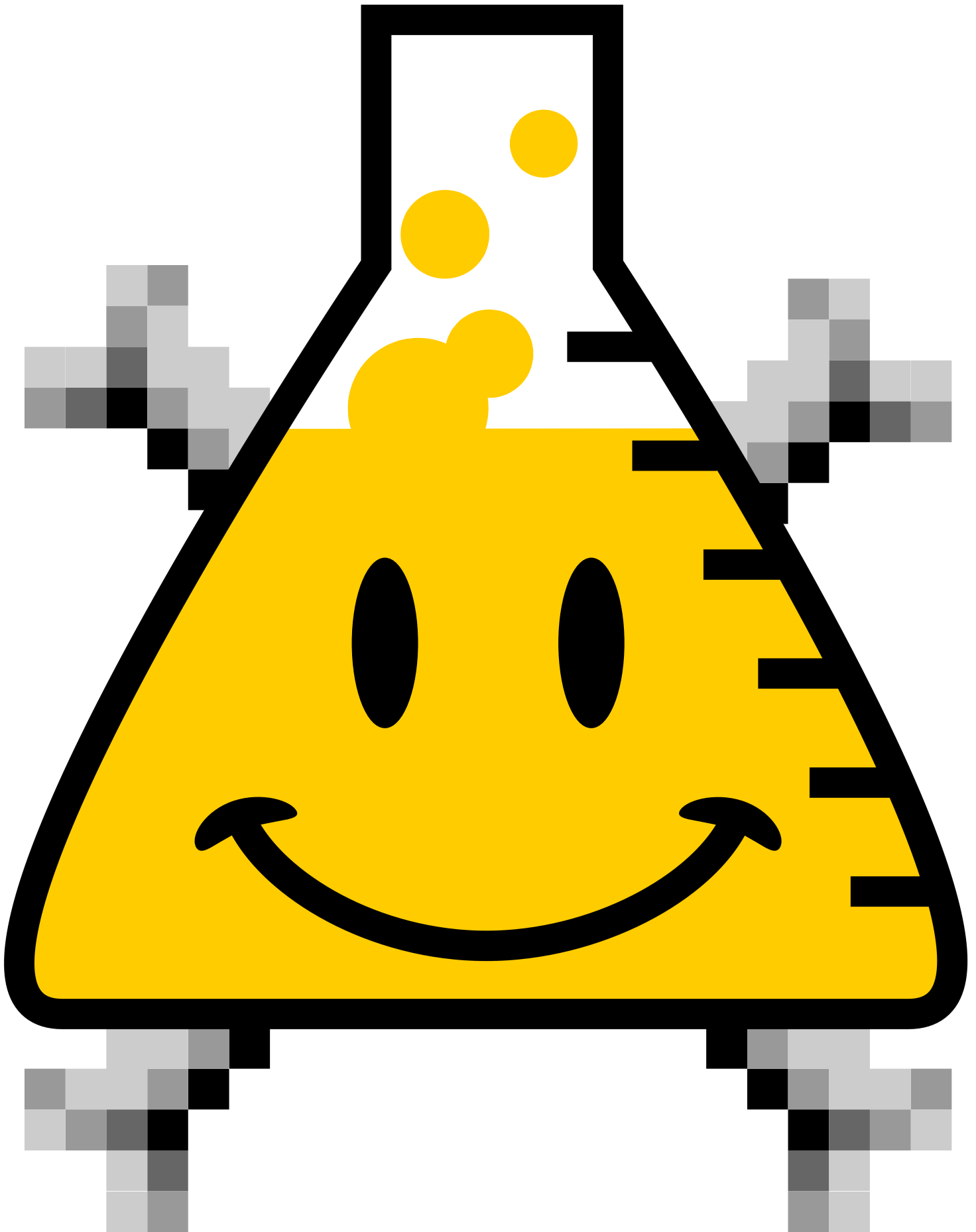
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